
Step 1: Connect Power BI to SQL Server Analysis Services (SSAS)

Open Power BI Desktop and navigate to **Home > Get Data > SQL Server Analysis Services database**. Enter the Server instance name (e.g., `localhost`) and specify the database. Select **Connect live** mode to query the Tabular model directly without importing data into memory.

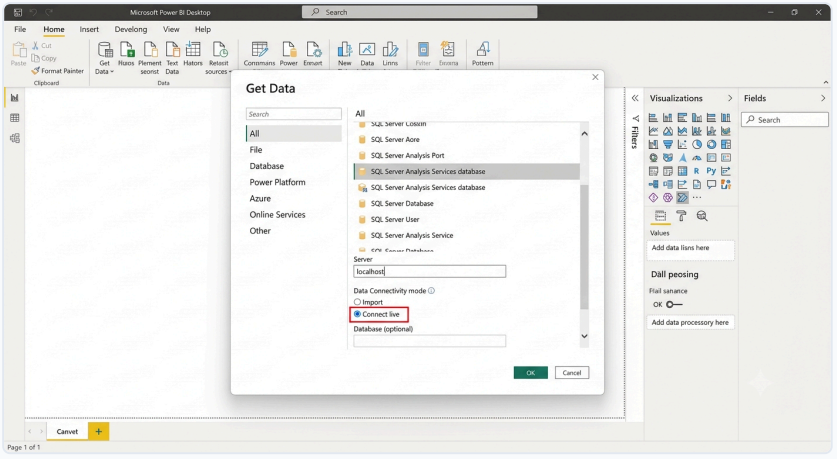


Figure 1.1: Connecting Power BI to SSAS in Live Connectivity Mode

Step 2: Verify Live Connection Mode & Field Navigator

Confirm the status bar displays *"Connected live to the Analysis Services database"* at bottom right. Observe that Data View and Model View transformation tools are disabled.

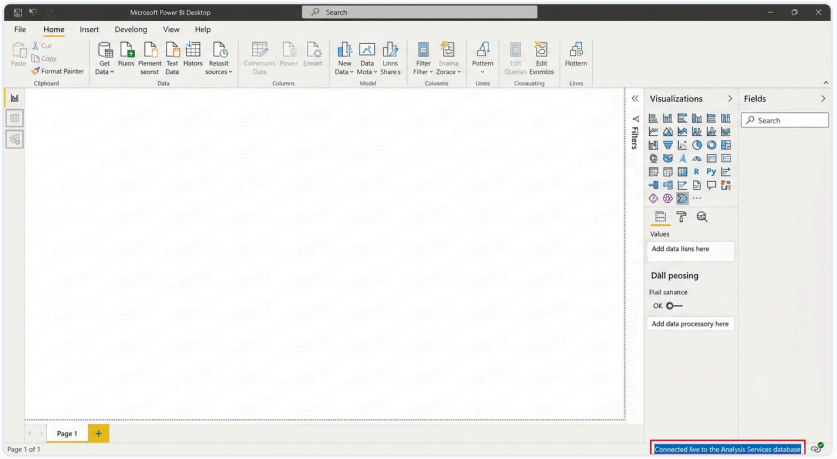


Figure 1.2: Status bar confirming active live connection and disabled transform tabs

Step 3: Define Static Row-Level Security (RLS) Roles

Navigate to **Modeling > Manage Roles**. Create static security roles and apply DAX filter expressions on target dimension tables to restrict user data access.

```
// Static RLS Filter Expression Department Security [Department] == "Sales"
```

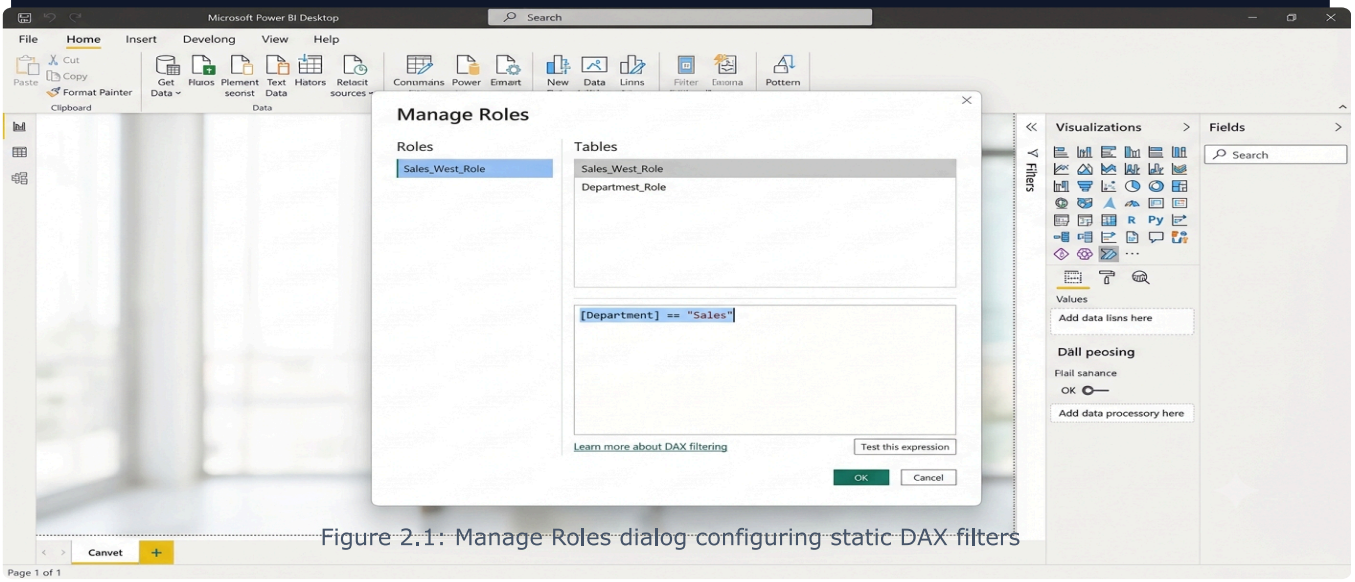


Figure 2.1: Manage Roles dialog configuring static DAX filters

Step 4: Configure Dynamic RLS Using USERPRINCIPALNAME()

Create a Dynamic Role using the native DAX security function `USERPRINCIPALNAME()` linked to a Security Mapping dimension table.

```
// Dynamic RLS DAX Filter on Security Table [UserPrincipalName] = USERPRINCIPALNAME()
```

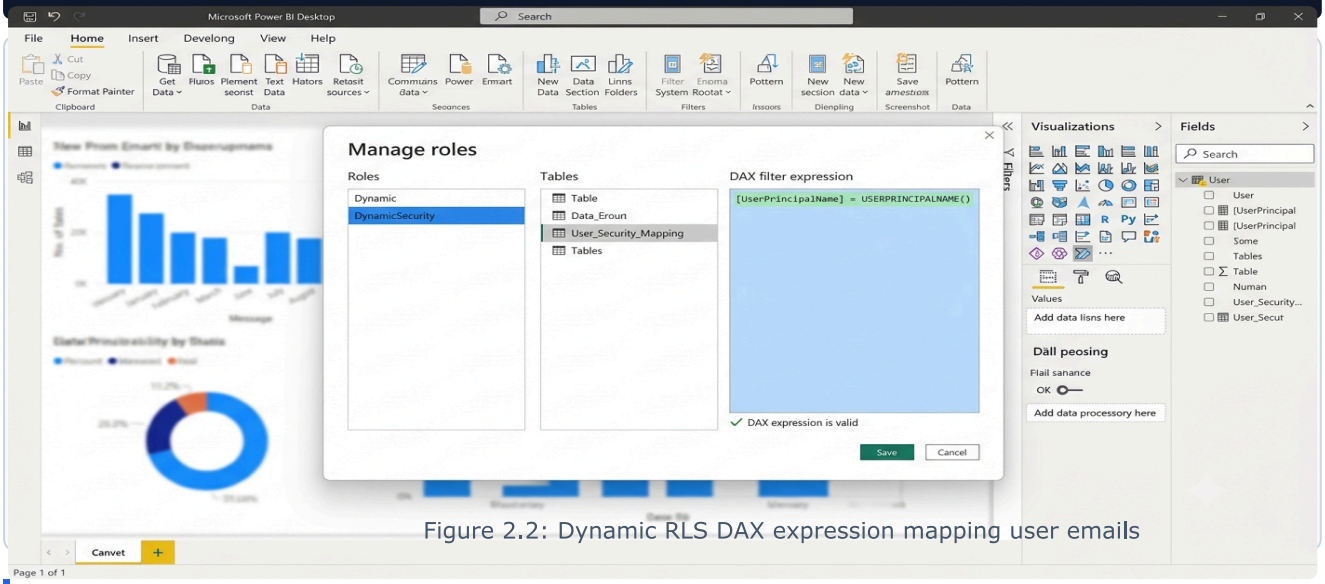
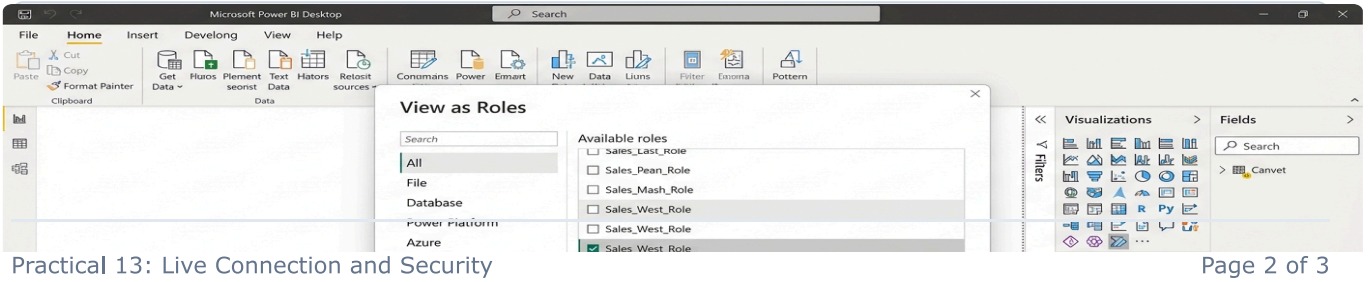


Figure 2.2: Dynamic RLS DAX expression mapping user emails

Step 5: Test Security Roles Using "View as Roles"

Select **Modeling > View as** in Power BI Desktop to simulate role execution locally and validate user security by entering specific test User Principal Names.



Step 6: Publish Report & Assign Security Members in Power BI Service

Publish the report file to the target Power BI Workspace. In Power BI Service, open dataset security settings (**Datasets > Security**) and add Azure AD users or Security Groups to their respective RLS roles.

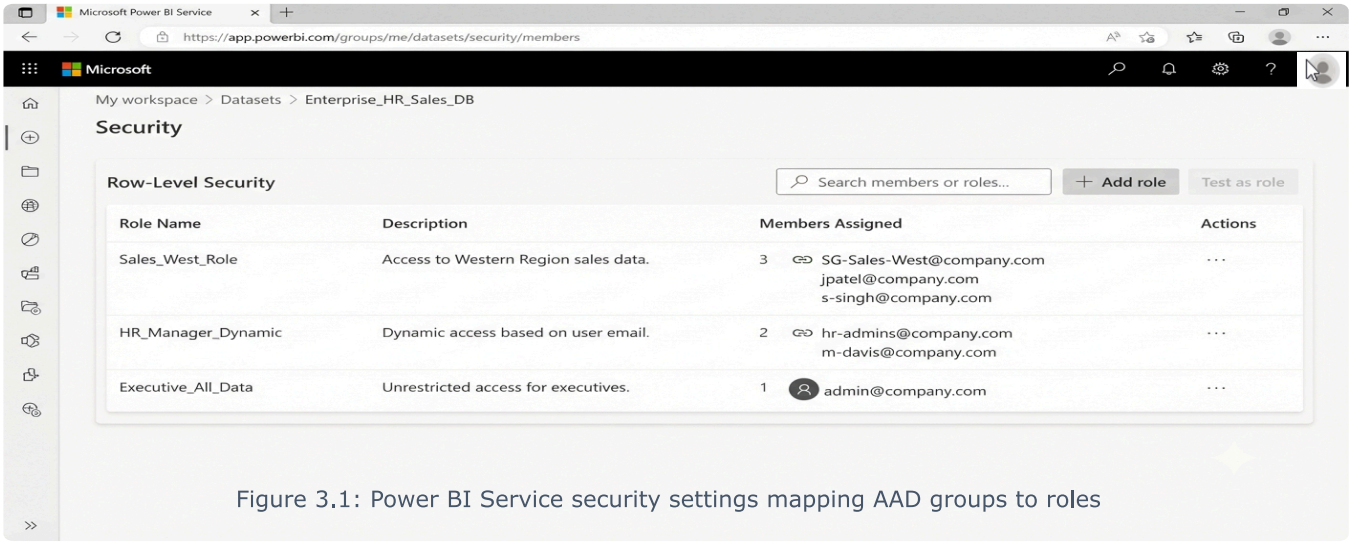


Figure 3.1: Power BI Service security settings mapping AAD groups to roles

Step 7: Verify Live Data Access & Row Filtering Integrity

Log in as different end-users in Power BI Service to verify that report visuals dynamically adapt to authorized subsets without data leaks. Save report as `Practical_13_SSAS_Live_RLS.pbix`.

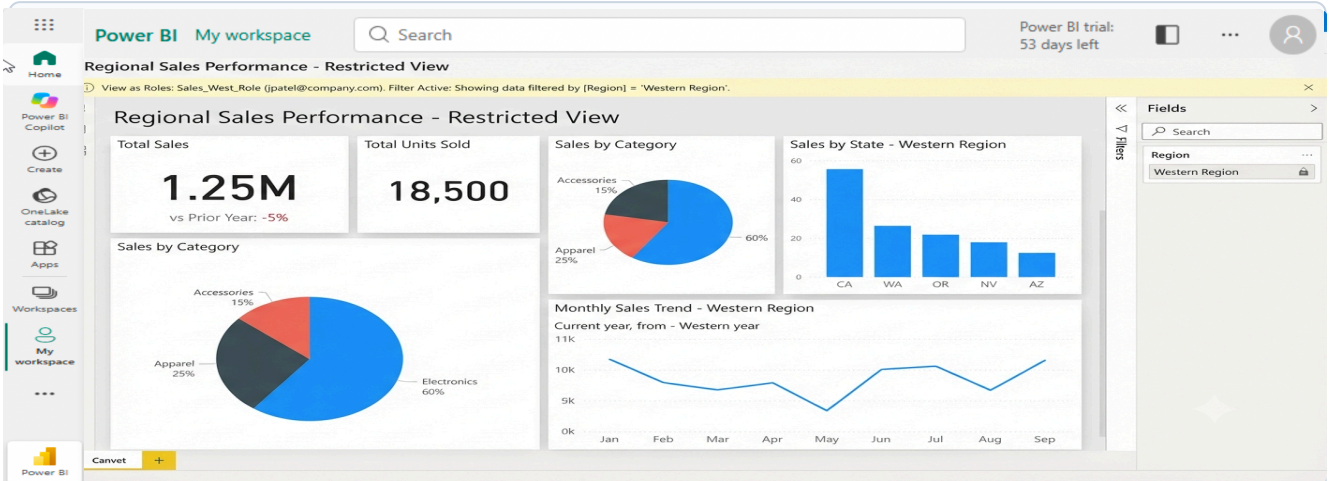


Figure 3.2: Verified live report visuals adhering to RLS filters

Result:

A live connection between Power BI Desktop and SQL Server Analysis Services (SSAS) was successfully established, retaining centralized governance and model performance. Both static and dynamic Row-Level Security (RLS) roles were configured using DAX functions (`USERPRINCIPALNAME()`) and validated.